KOMPAS

Focus: make the research novel by including the newest startups in relevant industries.

Circular Economy

Circular economy (CE) replaces linear economy (LE), defined by extraction, production, and disposal, with a cycle of resources. CE first gained urgency in 2015-2016, when research first identified the depletion risk of global resources. By 2017, concrete strategies for CE began to emerge. The global CE market is projected to grow from $149.86 billion in 2024 to $355.44 billion by 2032. The growing investment is attributable to several incentives: reduced dependence on extractive markets, savings on material costs, integration with ESG criteria, and the adoption of Product-as-a-Service (PaaS) business models.

The concept of CE has also evolved over the years. Since 2017, researchers have diverged on defining CE as narrow “3R” frameworks (Reduce, Reuse, Recycle) to broader “4R” or even multi-R models that emphasize “Rethink” and decoupling of economic growth from resource depletion. The focus on sustainable development has expanded from 11% to nearly 30% of research outputs, while attention to intergenerational equity and business models has also increased. These are promising for the future of CE.

Despite these advances, global circularity has paradoxically declined, from 9.1% in 2018 to 6.9% in 2025, according to the Circularity Gap Report. The foundation of CE is the Waste Management Hierarchy, prioritizing prevention, followed by reuse, recycling, energy recovery, and disposal as a last resort. Most of the $164 billion investments made from 2018 to 2023, for example, have concentrated on repair, resale, and recycling, which are reactive rather than proactive strategies. Instead, more effort should be concentrated on product design, material efficiency, and reuse.

In the startup landscape, innovation is emerging across design, reuse, and material recovery. Product design startups like The Paze develop modular, durable packaging with built-in recyclability, while upcycling ventures such as Insutex, MacroCycle Technologies, and Rethread Africa convert post-consumer waste into high-value materials. Rebaba reuses retired EV batteries as Battery Energy Storage Systems (BESS), extending product lifecycles beyond their primary use. Beyond reuse, waste restitution startups like C+UP transform captured CO_2 into propane, linking CE to CCUS, while Metallix invests $82M in metal extraction from industrial waste. Supply chain optimization firms like GoNina and Black Square leverage AI to map waste flows and reduce food or material losses. Together, these ventures show that CE’s next frontier lies not merely in recycling, but in redesigning systems to prevent waste generation in the first place.

Energy-Efficient Buildings

As of 2021, the building and construction sector accounts for 36% of global energy use and 37% of emissions, with developing regions such as Africa reaching as high as 60%. Energy-efficient buildings address this challenge through passive design, active design, energy-as-a-service (EaaS) models, and retrofitting.

Passive design minimizes energy demand by optimizing a building's physical form and materials to maintain comfortable indoor conditions with minimal mechanical intervention. It relies on a compact building form, super-insulated envelopes (including windows, walls, roofs, and doors), continuous air and vapor control layers, and balanced ventilation that recycles indoor heat. Passive systems are tailored to local climates. Thus, what works in cold, continental zones may fail in humid tropical regions. This makes climate suitability and adaptive standards essential. Though initial construction costs are higher, the long-term benefits include drastically lower energy consumption and improved occupant health. Yet, many countries still lack consistent standards for passive building certification, slowing widespread adoption and international scalability.

Active design, by contrast, focuses on optimizing operational energy use through automation and data-driven control. Smart buildings integrate high-efficiency HVAC systems, IoT sensors, and real-time analytics to automatically adjust lighting, heating, and ventilation based on occupancy and environmental conditions. Predictive maintenance algorithms detect when equipment will fail or lose efficiency, preventing unnecessary energy loss. By coordinating systems dynamically, active designs ensure that energy is only used when required, achieving savings of up to 30–50% in some commercial settings. Startups such as *Thermosphr* exemplify this approach, combining AI and thermodynamics to train HVAC systems for specific buildings, while other firms focus on integrating renewable energy sources like rooftop solar to power these systems directly.

The Energy-as-a-Service (EaaS) model transforms how building owners finance energy efficiency. Instead of bearing upfront capital costs, customers subscribe to an energy service, while providers design, finance, and operate the systems using their own capital. Customers then pay only for the energy consumed, sharing long-term savings with the service provider and creating a win-win situation. This approach aligns with growing corporate demand for low-carbon infrastructure without CAPEX burdens. The global EaaS market, valued at $70.63 billion in 2024, is projected to reach $151.27 billion by 2032. Companies such as *Schneider Electric*, *Unison Energy*, and *FESCO Energy* design and maintain microgrids and distributed power systems. EaaS exemplifies how financial innovation can complement technical progress, making clean energy upgrades accessible even in cost-sensitive markets.

Retrofitting, defined as upgrading existing structures with modern, efficient technologies, offers the greatest potential impact. According to the World Economic Forum, 75% of the buildings that will exist in 2050 are already standing today, underscoring the importance of revitalizing current infrastructure. Retrofitting can reduce building energy demand by up to 38%, translating into global emissions reductions greater than those achievable by the entire industrial or transport

sectors. It also yields economic and social co-benefits, such as reducing employee sickness by 20%, improving productivity by up to $7,500 per person annually, and creating 3.2 million new jobs each year. Innovative firms like *Aeroseal*, whose technology enables airtight duct sealing, *Inovues*, who specialize in window retrofits, and *View*, with their innovative dynamic glass, demonstrate how incremental upgrades can deliver transformative results. Policy frameworks such as the EU's Energy Performance of Buildings Directive (EPBD), enacted in 2024, further institutionalize this momentum, supporting small and medium-sized enterprises in Europe's construction industry while steering the global built environment toward a sustainable, low-carbon future.

Finally, we emphasize the lack of attention to passive buildings. While other aspects of energy efficient buildings have received hundreds of millions of funding, only several tens of millions have been invested into passive buildings. Since most buildings today will continue to exist decades after, it is an opportunity to optimize them through passive design or retrofitting.

Carbon Capture, Utilization, and Storage

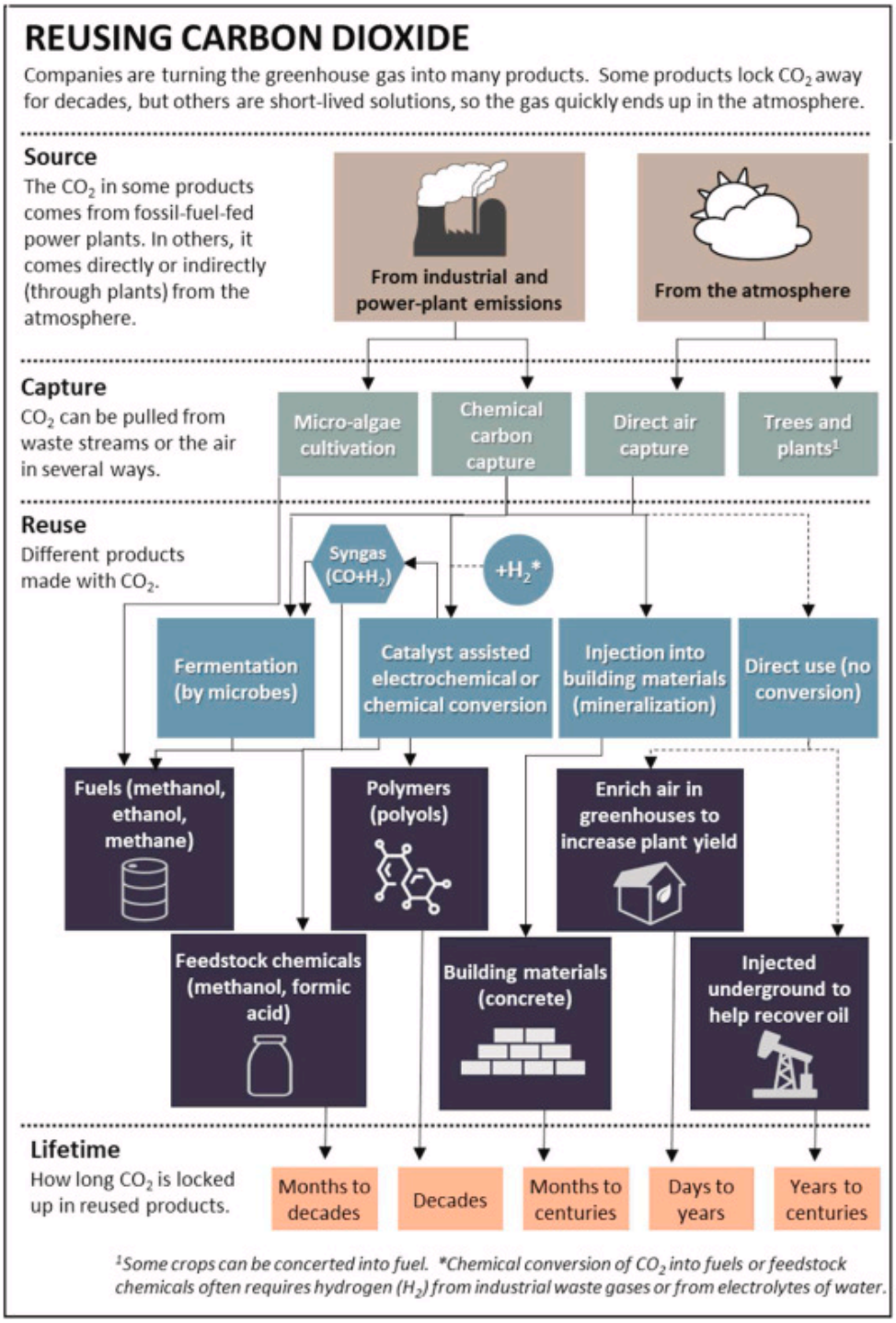
Carbon Capture, Utilization, and Storage (CCUS) refers to the process of capturing carbon dioxide (CO_2) produced during industrial activities and power generation, followed by its storage or reuse in other applications. It is widely viewed as a critical technology for addressing the imbalance between growing fossil fuel dependence and escalating greenhouse gas emissions, especially since over 84% of global energy demand is still met by fossil-based fuels. CO_2 concentration in the atmosphere has risen from 315 ppm to 415 ppm in just six decades, intensifying global warming through the trapping of infrared radiation. In 2018, 37.1 billion tons of CO_2 were emitted globally, including 5 billion tons from the United States, yet only 40 million tons were captured and stored annually as of 2019, which is a negligible fraction. The International Energy Agency (IEA) and Nordic Energy Research have described CCUS as "the most important option among new technologies for reducing industrial CO_2 emissions after 2030", though significant uncertainty remains regarding large-scale deployment. As the voluntary carbon market expands, CCUS is expected to represent a substantially larger share by 2030, reflecting its growing role in decarbonization.

The CCUS system consists of three main components: capture, utilization, and storage. CO_2 can be captured through indirect air capture by capturing high-concentration CO_2 near sources such as power plants or factories. Alternatively, it could be captured through direct air capture (DAC), which removes CO_2 from the ambient air, where concentrations are much lower. Indirect methods include post-combustion carbon capture, the most common in existing power plants; pre-combustion capture, used in industrial hydrogen or ammonia production; and oxyfuel combustion, which simplifies CO_2 separation by burning fuels in pure oxygen. Once captured, CO_2 must be compressed and transported through pipelines, ships, rail, or motor systems to its final destination for utilization or permanent storage. Utilization pathways are expanding beyond traditional enhanced oil recovery to include the production of fertilizer raw materials, biochar, fuel cells, bioproducts, chemicals, and polymers. However, the Intergovernmental Panel on Climate Change (IPCC) still identifies afforestation and forest restoration as the most

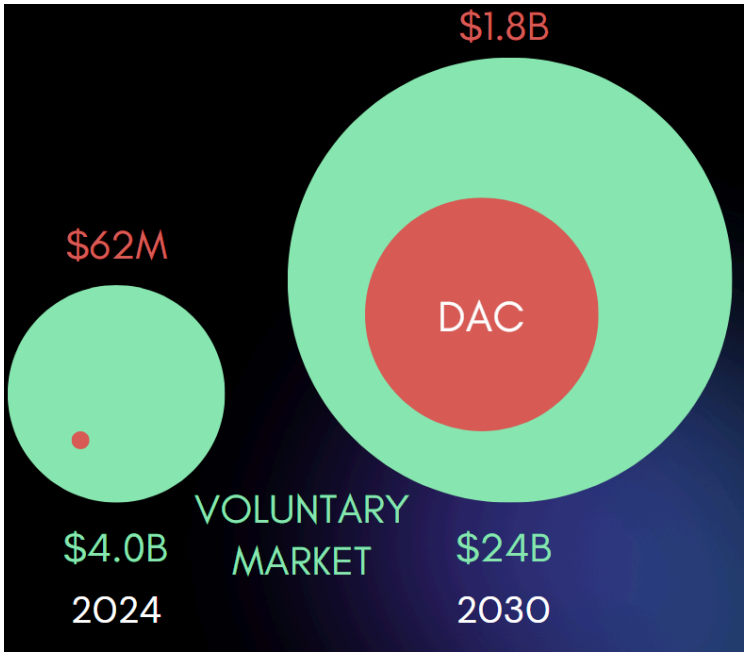
cost-effective carbon removal strategies, illustrating the diverse portfolio needed for achieving global climate goals.

Among these methods, Direct Air Capture (DAC) represents the fastest-growing and most transformative segment of CCUS, offering the potential for true negative emissions. Several emerging startups are driving cost reduction and scalability. Heirloom (Series B) utilizes a limestone-based mineral looping process, taking advantage of an abundant and inexpensive natural material. YAMA, still a pre-seed startup, applies a modular, hybrid electrical system that is highly scalable and could eventually achieve costs as low as $70 per ton of CO_2 at scale. Climeworks and Carbon Engineering, two of the largest players, employ solid sorbents and liquid chemical solutions to capture CO_2 directly from air. However, DAC systems currently face high costs of over $600 per ton driven by the dilute concentration of atmospheric CO_2 and the significant energy input required for capture and regeneration. Nevertheless, these costs are expected to fall below $200 per ton by the 2030s, aided by public and private investments. The U.S. Department of Energy (DOE) has committed $3.5 billion toward developing large-scale DAC Hubs capable of capturing at least 1 million tons of CO_2 per year, indicating strong policy support for technology maturation.

Economically, CCUS is closely tied to both the voluntary carbon market and government incentives. The voluntary market currently pays $600–$1,000 per ton of CO_2 removed, primarily to offset corporate reputational risks, while each carbon removal credit represents one verified ton of permanently removed CO_2, certified by independent auditors. In the United States, the 45Q tax credit provides $180 per ton for CO_2 that is captured and permanently stored underground, and $130 per ton for CO_2 that is utilized in industrial processes such as enhanced oil recovery. These policy measures, though impactful, remain limited in scale compared to the carbon tax systems adopted in other regions—Canada ($80/ton), the EU ($65–95/ton), and the UK ($65–75/ton). The absence of a national carbon tax in the United States remains a structural barrier, as such a mechanism would greatly strengthen economic incentives for widespread CCUS and DAC adoption.



Ways which CO2 could be captured and utilized. Source.



Increasing significance of direct air capture. Created by the author.

Circular Economy

https://www.sciencedirect.com/science/article/pii/S2666412721000167

- Linear economy (LE): extraction-production-disposal
- Progress:
    - 2015-2016, research realized that resources for production could be depleted quickly.
    - In 2017, research on ways and strategies for CE.
- Top-down approach should be implemented
    - EU ESPR
- Actual case studies required to assess feasibility of theory
- **$149.86 2024 → $355.44 billion by 2032**
- Benefits for a company:
    - Less reliant on global extractive markets
    - Savings on material costs
    - Product-as-a-Service (PaaS)
    - ESG criteria
- Waste Management Hierarchy: prevention (reducing waste at the source), followed by reuse, recycling, energy recovery, and finally, disposal as a last resort.

https://www.sciencedirect.com/science/article/pii/S0921344923001374

- Challenges:
    - Disagreement on the concept of CE which changed over time since 2017
        - individual R less, more 3R and 4R "R's" (Rethink, Reduce, Reuse, Recycle)
        - 11% to 27-31% for sustainable development as main goal
        - brand new concept: decoupling economic growth from resource depletion
        - 1% to 9-11%: concept of future generations
        - 11% to 21-27% business models
    - Recycle → Repair & Resale → better product design, in order of increasing impact. However most (100%-4.7%) of the $164 billion from 2018 to 2023 investments are into repair resale and recycling and not product design reuse and material efficiency (reactive, passive vs. proactive, active) https://cse-net.org/us-corporations-circular-economy/#:~:text=In addition%2C a recent Circle,%2C reuse%2C and material efficiency
    - Upfront costs for innovation
    - Paradox: The global circularity rate has steadily **fallen** from **9.1% in 2018** (when measurements began) to **6.9%** as of the latest analysis (**Circularity Gap Report 2025**).
- The past decade has seen a rapid increase in the number of CE studies, with focus areas becoming increasingly narrow in scope (industry-specific) and context (country-specific) – a potential threat to broader applicability.
- Consensus is important for policy making

Startups

- design products with “added value” (modularity, durability and longevity, disassembly)
    - The Paze (aero-flex packaging)
- Reusing the same product in different stages of its life
    - Upcycling: Insutex (textiles → insulation materials), MacroCycle Technologies (upcycles plastic with cyclic macromolecules), Rethread Africa (textiles → biomaterials for fashion)
    - Rebaba: EV batteries after retirement → Battery Energy Storage Systems (BESS)
    - As opposed to Downcycling (e.g. plastic bottle → carpets). This doesn’t inherently solve the problem that the product would eventually be thrown away.
- Restitution of solid waste
    - C+UP: Carbon dioxide → propane (related to CCUS)
    - Extracting metals from waste, $82M investment: https://metallix.com/
- managing the supply chain better
    - GoNina: Food waste app, managing demands and stocks for restaurants
    - Black Square: AI platform to map waste flows

Energy Efficient Buildings

Passive buildings https://jipr.springeropen.com/articles/10.1186/s43065-022-00068-z

- Globally as of 2021, buildings and building construction industry account for 36% and 37% of energy and emissions respectively. Asia (no china), central and south america: 23-25%. Africa: 60%.
- High initial upfront cost to design, but lower cost in the long term due to energy efficiency. Most contracts though prefer the opposite. However we must not “over-invest” into energy efficiency due to economical reasons.
- Climate suitability is very important since “passive”: tropicals vs cold climates
- many countries yet to have standards for passive buildings
- Technology: compact building form, super-insulated **building envelope (windows, roof, doors, walls)** with continuous control layers, balanced ventilation with minimal mechanical systems, high-performance windows, and air-tightness.

Active systems, smart buildings

- high efficiency HVAC
- automated systems (IoT) to control lighting, heating, cooling, and ventilation. When used in combination with real-time data (like occupancy levels), these systems ensure energy is only used when required. https://www.equans.com/glossary/what-are-energy-efficient-buildings#:~:text=Smart buildings also use automated,is only used when required
- data analytics to predict when equipment is likely to fail or become less efficient

On-site renewable energy such as solar panels.

energy-as-a-service models:

- No upfront cost! the customer subscribes, and the provider pays the upfront capital.
- Since the saving is long-term, the customer pays less of what it may have paid. Both parties benefit.
- Energy As A Service (EaaS) Market size was valued at USD 70.63 Billion in 2024 and is projected to reach USD 151.27 Billion by 2032 https://www.verifiedmarketresearch.com/product/energy-as-a-service-eaas-market/#:~:text=Energy As A Service (EaaS) Market Size And Forecast,10.05%25 from 2026 to 2032

Retrofitting

- “75% of the buildings that will be standing in 2050 already exist” (2024 edition of the World Economic Forum’s Transforming Energy Demand)
- buildings have the potential to reduce energy demand by 38% (12% globally), larger than industry and transportation.

- reducing staff sickness by 20%, improving employee productivity (up to $7,500 per person per year) and the creation of 3.2 million new jobs per year.

Policies

- EU EPBD, entered into force in 2024. efficient buildings creates jobs and enhance the competitiveness of the EU's construction industry (which is made up of small and medium-sized enterprises and employs almost 25 million people)

Startups

https://app.dealroom.co/lists/19530

- Paces
    - Software for Power and Data Center developers
    - de-risking
- EaaS
    - Schneider Electric (public company)
    - Unison Energy Unison Energy designs, builds, owns, operates, and maintains microgrids for the long term. We finance the projects using our own capital, so you pay only for the energy consumed.
    - FESCO energy: no CAPEX for energy transition for industries
- Retrofitting
    - Aeroseal achieves airtight sealing with Aeroseal Duct
    - Inovues provides smart, energy-efficient windows with its tech, saving up to 40% of energy
    - View provides dynamic windows
- On-site renewable energy (biggest aspect with $4.7B funding): GreenLancer, Perfecta Energia, Vespr Solar installs solar panels cheaply and quickly
- Active systems:
    - Thermosphr uses AI and thermodynamics to control HVAC after training for specific building
- Passive (only $30M funding as of today)
    - Naava provides green furniture
    - Skycool provides radiative cooling material for HVAC to radiate heat into the sky and improve efficiency
- Sustainable cement, steel, building with blocks made from recycled plastic

Challenges

- Extreme weather
- Risk (profit only in the long term)

CCUS

https://www.sciencedirect.com/science/article/pii/S1364032123000710

https://www.sciencedirect.com/science/article/pii/S2213343723021322

- Carbon capture, utilization, and sequestration/storage (CCUS) is the process of capturing CO2, i.e., produced during industrial processes and power generation, followed by its storage.\
    - CCUS would make up a significantly more portion of the voluntary carbon market by 2030.
    - the International Energy Agency and Nordic Energy Research have even claimed that CCUS "represents the most important option among new technologies for reducing industrial CO2 emissions after 2030. Currently, great uncertainties exist as to how to deploy CCS …"
        - There are multiple ways to capture and reuse CO2, as shown in the figure. https://ars.els-cdn.com/content/image/1-s2.0-S1364032123000710-gr1.jpg
    - more than 84% of global energy demand is met by fossil-based fuels
    - The CO2 concentration, i.e., mainly responsible for global warming due to its infrared rays trapping nature, has escalated from 315 parts/million to 415 parts/million in just the past six decades
    - In 2019, 40 million metric tons of CO2 are captured and stored globally each year
    - Globally, 37.1 billion tons of CO2 was released into the atmosphere in 2018, with 5 billion from the US
- Aspects:
    - direct air capture: from ambient air where concentration is low
        - YAMA (pre-seed): modular system, scalable, hybrid electrical approach, by far the most scalable and economically feasible approach, could reach $70/tCO2 at scale.
        - Heirloom: Series B, uses limestone (cheap material, Mineral Loop approach)
        - Climeworks, Carbon Engineering: one of the largest companies, uses solid sorbent
    - indirect air capture: installed in factories to capture high concentration CO2
        - post-combustion carbon capture (i.e., the primary method employed in the existing power plants)
        - precombustion carbon capture (i.e., highly used in industrial processes)
        - oxyfuel combustion systems
    - transportation via pipelines, ships, rail, and motor transport
    - utilization pathways:
        - fertilizer raw materials, enhanced oil recovery, fuel cells, and biochar. bioproducts, chemicals, and polymers.

    - According to the Intergovernmental Panel on Climate Change, the easiest, most economical, and the most effective way is the restoration of the forests by afforestation.

DAC, voluntary market and carbon credits

- we emit over 35 billion ton CO2 / year, need to remove at least several billion per year to reach 2050 climate goal
- US 45Q tax credit: $180 (if stored underground), $130 t/CO2 (if utilized e.g. through oil recovery) → carbon removers
- **Voluntary market**: pays $600~$1000 t/CO2 in place of reputational damage
- To keep global temperature rise below 1.5 degrees celsius, we'll need to remove **billions of tons of CO2** each year – a volume equivalent to 10–20% of today's annual global emissions.

Capital and Policy

Each removal credit reflects exactly one ton of CO2 that has been permanently removed from the atmosphere, and is guaranteed by a third party that independently monitors our technology.

Today DAC is expensive (>$600/t). Companies paying that premium help fund cost reductions so prices can fall below $200/t in the 2030s.

- Key requirement: captured CO_2 must be compressed, transported, and permanently stored (e.g. underground, mineralized) or utilized (e.g. in concrete, fuels).
- DOE has put $3.5B into DAC Hubs (large plants ≥1 Mt/yr) to accelerate scaling

Carbon tax:

- Canada: $80
- EU: $65-95
- UK: $65-75
- US: NO CARBON TAX. There would be greater incentive for DAC if there were carbon tax in the US.